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Antigen presentation to naive CD4 T cells in the lymph node

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Although the presentation of peptide–major histocompatibility complex class II (pMHC class II) complexes to CD4 T cells has been studied extensively *in vitro*, knowledge of this process *in vivo* is limited. Unlike the *in vitro* situation, antigen presentation *in vivo* takes place within a complex microenvironment in which the movements of antigens, antigen-presenting cells (APCs) and T cells are governed by anatomic constraints. Here we review developments in the areas of lymph node architecture, APC subsets and T cell activation that have shed light on how antigen presentation occurs in the lymph nodes.

Initiation of many aspects of the adaptive immune response depends on CD4 T cell recognition of foreign pMHC class II complexes on the surface of APCs. Although several cell types express MHC class II molecules, it is believed that dendritic cells (DCs) are the initiating APCs, because they are located at portals of antigen entry and are efficient at antigen uptake, processing, pMHC complex display and migration to the T cells areas of secondary lymphoid organs[1]. Moreover, the activation state of the DCs at the time of initial pMHC complex presentation may determine whether the responding naive T cells become memory cells or are deleted, silenced or suppressed[2,3].

Although much circumstantial evidence indicates the DC is essential, it is still not clear which cells first present pMHC class II complexes to naive CD4 T cells *in vivo*. It is clear from *in vitro* experiments that other MHC class II–expressing cells (for example, macrophages and B cells) are capable of presenting pMHC class II complexes and stimulating naive CD4 T cells, albeit less efficiently than DCs[4,5]. Given this potential, could macrophages or B cells initiate the primary CD4 T cell response? In addition, if DCs are the initiating APCs, which of the many subsets are involved? Is the response initiated exclusively by DCs that migrate from the nonlymphoid site of antigen entry, or are the DCs that are already in the secondary lymphoid organs involved? If so, do migrating and resident DCs induce different types of T cell responses?

The blood and lymphatic vasculature that pervades the site at which antigen enters the body determines, to a large extent, the type of secondary lymphoid organ in which the CD4 T cell response begins[6]. This 'plumbing' determines that immune responses to antigens, which enter the body through the skin, mucosal surfaces or blood, begin in the peripheral lymph nodes, mucosal lymphoid organs or spleen, respectively. This review will focus on pMHC class II complex presentation to naive CD4 T cells in lymph nodes.

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CD4 T cell activation in secondary lymphoid organs

In many cases, antigens that are eventually presented in the lymph nodes first enter the body in nonlymphoid tissues, such as the subcutaneous tissue after skin puncture. It is therefore theoretically possible that nonlymphoid sites of antigen entry are the places where naive CD4 T cells are first activated. This is unlikely, as much evidence indicates that naive CD4 T cells have little, if any, access to parts of the body other than the blood and secondary lymphoid organs[7]. For example, naive CD4 T cells could only be detected in the lymph nodes, spleen and mucosal lymphoid tissues using whole-animal immunohistology[8]. This anatomic limitation indicates that in most cases the initial activation of naive CD4 T cells must occur in secondary lymphoid organs as a result of antigen transport from nonlymphoid tissues. As noted below, however, the lungs may be an exception to this rule.

MHC class II–expressing cells in the lymph node

Three cell types within the murine lymph node express MHC class II molecules, and thus have the potential to activate CD4 T cells: macrophages, B cells and DCs. However, because macrophages are located mainly in nonlymphoid tissues, the red pulp of the spleen, and the subcapsular and medullary sinuses of lymph nodes[9,10], but not in the T cell areas, it is unlikely that these cells act as initial APCs. Indeed, T cell–dependent immune responses are induced normally in mice deficient in macrophage colony-stimulating factor that lack sinusoidal macrophages[11]. Although this result indicates that macrophages are not essential for initiation of the CD4 T cell response, not all macrophages are absent in mice deficient in macrophage colony-stimulating factor[11]. Thus, it remains possible that a subset of macrophages is involved in initial pMHC class II complex presentation to naive CD4 T cells in the lymph nodes. However, the abundance of macrophages in many nonlymphoid organs indicates that these cells are important APCs for effector CD4 T cells that migrate into nonlymphoid organs at late stages of the primary immune response[8].

B cells are also unlikely to be involved in the initial pMHC class II complex presentation to naive CD4 T cells. Although it was reported that all B cells in the lymph nodes produce lysozyme pMHC class II

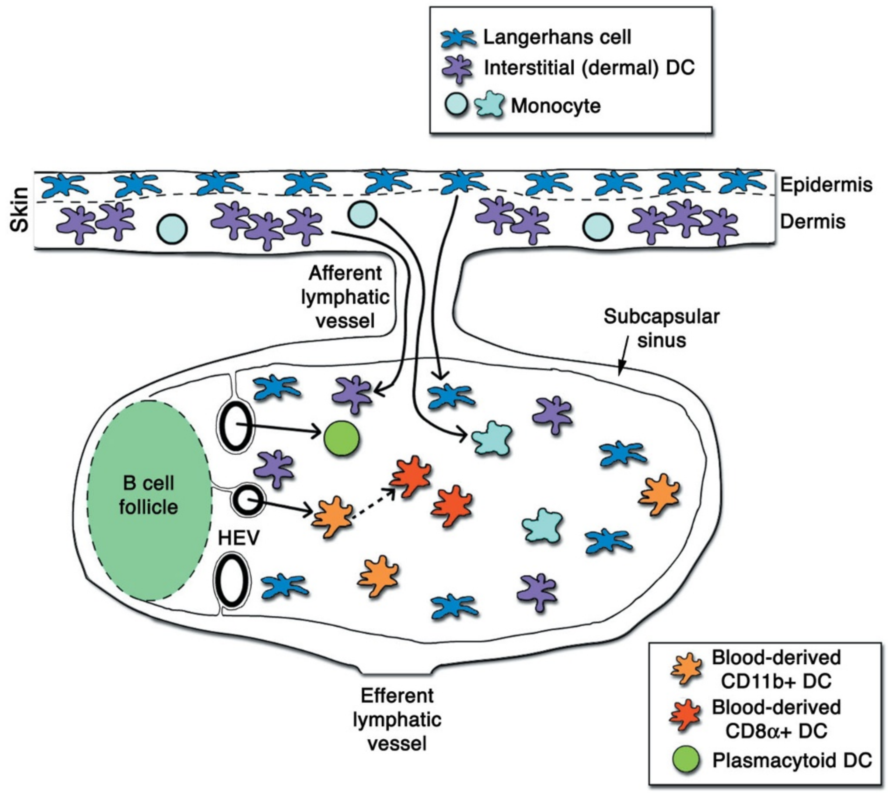

Figure 1 Location of DC subsets in the lymph node. The skin contains two main types of immature DCs: Langerhans cells (dark blue) in the epidermis, and interstitial DCs (purple) in the dermis. The dermal DCs are probably similar in origin and function to interstitial DCs that exist in other nonlymphoid tissues, whereas Langerhans cells are unique to the epidermis. Both populations of DCs migrate to the draining lymph node through afferent lymphatic vessels at a low constant rate, which increases considerably in response to inflammatory signals. Inflamed skin also contains a population of monocytes (light blue), which, after migrating to the draining lymph node, can acquire a DC phenotype and morphology. In addition to these three skin-derived DC subsets, a skin-draining lymph node contains three DC subsets that enter through the HEV: blood-derived CD11b$^+$ (orange), blood-derived CD8α^+ (red) and plasmacytoid (green) DCs. Blood-derived CD8α^+ DCs localize to the center of the cortex, whereas blood-derived CD11b$^+$ DCs, Langerhans cell immigrants and dermal DC immigrants occupy the outer cortex, near the conduits that surround the HEVs and follicles.

complexes several hours after injection of a large amount of lysozyme[12], earlier work indicated that B cells lacking an antigen-specific surface immunoglobulin (Ig) molecule are very inefficient at taking up antigen and stimulating relevant CD4 T cells[13]. Indeed, in a different study in which less antigen was injected, less than 1% of the B cells in the draining lymph nodes produced pMHC class II complexes[14]. In any case, no matter how many B cells produce pMHC class II complexes from a given antigen, the paucity of B cells in the T cell areas indicates that B cells are unlikely to be the initial APCs for naive T cells in most situations. In support of this contention, *in vivo* proliferation of antigen-specific CD4 T cells after immunization is normal in B cell–deficient mice[15,16].

Although these results indicate that B cells are not required, they may still participate as initiating APCs for naive CD4 T cells. In support of this idea, naive antigen-specific CD4 T cells proliferated in mice injected with antigen-pulsed B cells expressing surface immunoglobulin specific for the antigen[17]. The T cells were stimulated in part by direct recognition of pMHC class II complexes on the injected B cells. The antigen-specific CD4 T cells and B cells could have interacted as the B cells traveled out of the blood and through the T cell areas on the way to the B cell–rich follicles[18]. Direct initial presentation by B cells, however, may be limited to this special situation in which large numbers of B cells display pMHC class II complexes. Most of the literature on soluble antigens indicates that rare antigen-specific B cells only present pMHC class II complexes to CD4 T cells that were initially activated by DCs[19,20] (**Supplementary Video** online, 00:00:37–00:00:45).

As B cells and macrophages are not required for activation of naive CD4 T cells, and DCs are the only other MHC class II–expressing cells that occupy the T cell areas of the lymph nodes[10,21], DCs are probably the essential APCs for initial pMHC class II presentation to naive CD4 T cells in most situations, although they may also be involved in the activation of B cells[22] and natural killer cells[23]. In addition to this indirect evidence, direct confirmation that DCs are the main initiating APCs comes from two studies showing interactions between antigen-bearing DCs and antigen-specific CD4 T cells *in situ*. One showed that ovalbumin-specific T cells formed small clusters around fluorescent DCs in draining lymph nodes 24 hours after subcutaneous injection of fluorochrome-labeled ovalbumin[24]. Similarly, another showed that DCs were the main cell type to engulf fluorescent latex beads coated with ovalbumin, and that ovalbumin-specific T cells were found adjacent to bead-containing DCs in peribronchial lymph nodes 12 hours after beads were given intranasally[25]. In addition, real-time microscopic imaging showed that antigen-specific T cells form clusters 24 hours after exposure to subcutaneously injected soluble antigen[26]. Although it was not demonstrated in the last study that DCs were in these T cell clusters, the location of the clusters in the T cell areas was consistent with the possibility that DCs were present. Indeed, a separate *ex vivo* analysis of these clusters showed that they contain DCs[27].

These are important experiments, as they attempted to detect antigen presentation *in situ*. However, this approach has its limitations. By tracking intact antigen, it can only be assumed that the cells that contained the antigen and were interacting with antigen-specific T cells actually displayed the relevant pMHC complexes. Also, these experiments did not demonstrate that the DC–T cell interactions caused the interacting T cells to become activated. However, together with the studies of B cell– and macrophage-deficient mice, these physical findings provide evidence that DCs directly interact with naive T cells, and that because these interactions are antigen-dependent, they probably represent actual antigen presentation events.

Given that DCs are probably the cells that first present pMHC class II complexes to naive CD4 T cells, the question then becomes which of the many DC subsets are involved. The answer to this question seems to depend on the form of the antigen and the way that it enters the body. The following sections summarize what is known about pMHC class II complex presentation by individual DC subsets.

Antigen presentation by blood-derived DCs

DCs are large, stellate bone marrow–derived cells that express MHC class II molecules constitutively and are found in nonlymphoid tissues and the T cell areas of secondary lymphoid organs[1]. In the mouse, the CD11c integrin seems to mark most DCs[1], although it is also expressed on monocytes and a subset of 'antigen-experienced' CD8 T cells[28]. As many as six different subsets of DCs occupy the lymph nodes (**Fig. 1**). Some subsets are derived from a precursor that enters the lymph node from the blood[29], whereas others are derived from precursors that migrate from the tissue to the lymph node through the afferent lymphatic vessels[30,31]. These subtypes and their precursors have been reviewed[32].

Three of the six types of DCs are found in all secondary lymphoid organs, including those that lack afferent lymphatic vessels, indicating that these cells arrive in the lymphoid organs from the blood. One type of blood-derived DC expresses the myeloid marker CD11b, may express CD4, does not express CD8α or the CD205 integrin, and is often referred to as the myeloid DC[33]. However, these cells will be referred to here as blood-derived CD11b$^+$ DCs, because it is not clear that they are of myeloid lineage[34]. A second type lacks CD11b but expresses CD8α and CD205, and is often referred to as the lymphoid DC[33]. These cells will be referred to here as blood-derived CD8α$^+$ DCs, because it is not clear that they are of lymphoid lineage[34]. Although one study suggested that blood-derived CD11b$^+$ DCs differentiate into blood-derived CD8α$^+$ DCs after adoptive transfer, indicating that the former population gives rise to the latter[35], another suggested that this is not the case[36]. Lymph nodes also contain a population of CD11c$^+$CD11b$^-$ cells that express the CD45 isoform (B220) normally expressed by B cells and have a plasmacytoid morphology[37]. These cells will be referred to here as plasmacytoid DCs.

Evidence that blood-derived CD11b$^+$ DCs present antigens to naive CD4 T cells in the lymph nodes comes from work showing that ovalbumin-specific T cells interact with CD11b$^+$ but not CD205$^+$ DCs in draining lymph nodes 18 hours after subcutaneous injection of fluorochrome-labeled ovalbumin[38]. Similarly, CD11b$^+$CD8α$^-$ DCs produce pMHC class II complexes from an inhaled Leishmania protein[39]. The CD11b$^+$CD8α$^-$CD205$^-$ phenotype is consistent with the possibility that the cells identified in these studies were blood-derived CD11b$^+$ DCs, although other evidence (discussed below) indicates that they may have been tissue-derived interstitial (dermal) DC immigrants, which are CD11b$^+$CD8α$^-$CD205^{lo}.

Several studies have shown a principal function for blood-derived CD8α$^+$ DCs in the presentation of pMHC class II complexes derived from particulate antigens: B cell–associated antigen in one case[18], and bacteria in another[40]. In contrast, blood-derived CD8α$^+$ DCs produce pMHC class II complexes from injected soluble antigens less efficiently than do blood-derived CD11b$^+$ DCs, or not at all[38,41,42]. This could be explained by a high rate of phagocytosis and a low rate of pinocytosis. Although one study reported that blood-derived CD8α$^+$ DCs produce pMHC class II complexes from a soluble antigen[43], as noted below, these CD8α$^+$ DCs could have been derived from Langerhans cells, which also express CD8α, albeit at an intermediate level[44].

Plasmacytoid DCs express very low amounts of MHC class II molecules in resting conditions[37]. However, they increase surface expression of these molecules after stimulation with a Toll-like receptor ligand and produce large amounts of interferon-α[45]. As plasmacytoid DCs are missing from the lymph nodes of CD62L-deficient mice[37], it is likely that they enter lymph nodes from the blood by passing through the high endothelial venules (HEVs). Although plasmacytoid DCs are capable of presenting pMHC complexes to T cells *in vitro*[46–48], their function as APCs *in vivo* has not been established.

Antigen presentation by tissue-derived DCs

Three additional DC subsets are found in lymph nodes, but not in the spleen. Because the spleen lacks afferent lymphatic drainage, these cells must be migrants that move into the lymph node from interstitial tissue through afferent lymphatic vessels. One of these populations consists of MHC class II^{hi}, CD11c$^+$ cells that express CD11b and lack CD8α and are thus similar to blood-derived CD11b$^+$ DCs, but also express small amounts of CD205[30,31,33,44,49]. These cells will be referred to here as interstitial DC immigrants, or in the case of skin, dermal DC immigrants. The lymph nodes that drain the skin contain another population of MHC class II^{hi} DCs that express high amounts of CD205, low amounts of CD11b and intermediate amounts of CD8α. As these cells are found at very low frequency in non-skin-draining lymph nodes or the spleen[30] and they express the Langerhans cell–specific protein Langerin[44], it is likely that they are derived from Langerhans cells that migrated from the skin through the afferent lymph. We will refer to these cells as Langerhans cell immigrants.

Langerhans cells and interstitial DCs migrate continually to the draining lymph nodes through the afferent lymph at a low rate[50], but this increases in response to tissue inflammation[51]. A third subset of CD11c^{lo} DCs, derived from monocytes that enter inflamed tissues, can also appear transiently in the lymph node[52].

A rich body of literature suggests that Langerhans cells are involved in the presentation of antigens that enter the body through the epidermis[1,53–55]. This contention is based largely on experiments done with reactive haptens, which, when applied to the skin surface, covalently attach to proteins within the epidermis. Langerhans cells efficiently take up hapten-labeled proteins and produce haptenated pMHC class II complexes from these proteins through the exogenous antigen processing pathway[56]. Chemically reactive haptens also stimulate the production of proinflammatory cytokines such as interleukin-1 (IL-1) and tumor necrosis factor-α[57], probably by causing cell damage. These cytokines increase the rate at which Langerhans cells leave the epidermis[58], enter local afferent lymphatic vessels and migrate into the T cell areas of the draining lymph nodes, beginning about 12 hours and peaking 3–4 days after application of the hapten[30,50,59]. Early evidence that Langerhans cells are involved in antigen presentation *in vivo* was that allogeneic tissues that lack Langerhans cells do not induce a delayed-type hypersensitivity response[60].

The ability of Langerhans cells to stimulate the T cell response to a subcutaneous tumor has also been demonstrated[61]. In this study, polymer rods containing ovalbumin alone or ovalbumin and CCL19, a DC-attracting chemokine, were implanted under the skin of mice. Only mice exposed to both CCL19 and ovalbumin rejected ovalbumin-transduced tumor cells *in vivo*, indicating that the recruited Langerhans cells were important for inducing protective immunity.

Finally, it was shown that MHC class II^{hi} tissue-derived DCs, consisting of both Langerhans cell immigrants and dermal DC immigrants, are the first cells to present pMHC class II complexes to naive CD4 T cells after a subcutaneous injection of a soluble antigen[14]. This study used the Y-A*e* monoclonal antibody, which is specific for a complex composed of I-Eα peptide 52-68 (pEα) bound to the I-A^b MHC class II molecule[62–64], to identify cells in the lymph nodes that displayed pMHC class II complexes derived from a soluble recombinant protein containing pEα. By 4 hours after injection, most Langerhans cell immigrants and dermal DC immigrants, but not blood-derived CD11b$^+$ or CD8α$^+$ DCs, had generated pMHC class II complexes

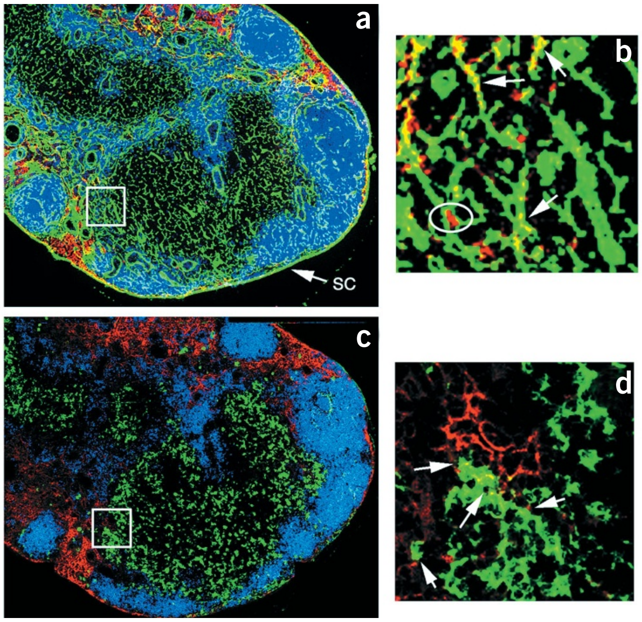

Figure 2 Localization of lymph node DCs, injected soluble antigen and collagen network *in situ*. Small soluble proteins introduced into tissue travel rapidly to the draining lymph node through the afferent lymphatic vessels. The proteins first enter the subcapsular sinus of the lymph node, and then enter the conduits that connect the subcapsular sinus to HEVs that border the B cell-rich follicles. In each of two serial sections, a fluorescent protein (red) is found in the conduits of the draining lymph node 30 minutes after injection. B cell–rich follicles are shown in blue (stained with antibody to B220). (**a**) Conduits containing collagen type III (green) fibers link the subcapsular sinus (SC) to the HEVs (round or elliptical objects), and are densest near the B cell-rich follicles. At 30 minutes after subcutaneous injection, soluble antigen (red) co-localizes with the subcapsular sinus and conduits nearest the subcapsular sinus. (**b**) Higher magnification of the area outlined by the white square in **a** shows that some of the soluble protein colocalizes with the conduits (yellow; arrows), whereas some antigen is located just outside of but is associated with the conduits (red; circle). (**c**) DCs expressing large amounts of CD40 (green) near the follicles may have access to soluble antigen (red) that is carried in the conduits that are abundant in this area. (**d**) Higher magnification of the area outlined by the white square in **c** shows CD40^{hi} DCs containing antigen (yellow; arrows).

from the antigen. This early display of pMHC class II complexes coincided with CD69 induction on, and IL-2 production by, antigen-specific CD4 T cells.

Analysis of DC subsets in the skin-draining nodes can be complicated by the similar surface phenotype of Langerhans cell immigrants and dermal DC immigrants. Both populations express similar amounts of CD11c, MHC class II and CD40, and can be distinguished only by subtle differences in the amounts of CD8α, CD11b and CD205. Thus, the MHC class II^{hi} populations of DCs shown to cause tumor rejection[61] or delayed-type hypersensitivity reactions[60] could have included dermal DCs. In the future, it will be necessary to use Langerhans cell–specific markers such as Langerin[59,65] to unambiguously determine which skin-derived DC subsets are involved in antigen presentation.

Interstitial DC immigrants in lymph nodes are similar to Langerhans cell immigrants except that their immature forms are located in interstitial spaces other than the epidermis. The involvement of interstitial DC immigrants as APCs can be inferred from several recent reports. As described above, *in situ* pMHC class II complex display by interstitial (dermal) DC immigrants was directly detected using the Y-A*e* antibody after subcutaneous injection of a soluble antigen containing pEα[14]. In addition, CD11c$^+$, MHC^{hi} cells containing fluorochrome-labeled ovalbumin appeared in the lung-associated lymph nodes after intratracheal administration of this antigen[66]. As these cells expressed CD11b and CD205, not CD8α, they were probably interstitial DCs that acquired antigen in the interstitial tissue of the trachea and then migrated to the draining lymph nodes. When isolated from the peribronchial lymph nodes, these DCs stimulated ovalbumin-specific CD4 T cells *in vitro*[66], indicating that the cells displayed the relevant pMHC class II complexes *in vivo*. However, T cell priming in these conditions occurs in mice lacking lymph nodes, raising the possibility that the initial pMHC class II presentation takes place in the lung parenchyma[67], not the secondary lymphoid organs as in most other immune responses.

Interstitial DC immigrants may also be the initiating APCs for antigens that enter the body through the reproductive tract. Interstitial DC immigrants from the submucosa, but not Langerhans cell immigrants or blood-derived CD8α$^+$ DCs, stimulated herpes simplex virus 2–specific CD4 T cells after intravaginal inoculation of herpes simplex virus 2 (ref. 68).

Subcutaneously injected fluorescent microbeads are engulfed not only by interstitial DCs but also by monocytes that quickly enter the injection site from the blood, probably in response to inflammatory mediators induced by tissue damage caused by the injection[52]. Within 24 hours, cells containing more than one bead and expressing large amounts of MHC class I and II molecules and some DC-specific molecules, but small amounts of CD11c and no Langerhans cell markers, appear in the draining lymph nodes. Monocytes that engulf particles and transmigrate *in vitro* across an artificial endothelial cell layer also acquire phenotypic characteristics of DCs[69]. These results indicate that monocytes that enter tissue sites and engulf particles receive signals to differentiate into DCs in the process of migrating from the tissue into an afferent lymphatic vessel. Although these monocyte-derived DCs are able to stimulate naive T cells *in vitro*[70], their function in antigen presentation *in vivo* has yet to be determined.

Migrating and resident DCs

In vitro studies have shown that proinflammatory cytokines cause Langerhans cells and other types of immature DCs to express more costimulatory molecules, reduce antigen uptake and processing, and stabilize surface pMHC class II complexes[1]. This process, often referred to as DC maturation, is thought to be coupled with DC migration, as inflammatory stimuli also induce tissue-resident DCs to travel to the draining lymph nodes. These linked processes would ensure that DCs migrating from the tissue are prepared to stimulate the appropriate naive T cells after arriving in the draining lymph node. Based on this model, most of the tissue-derived DCs that reside in the lymph nodes should have the properties of mature DCs. Indeed, Langerin-positive cells from skin-draining lymph nodes have large amounts of MHC class II molecules, and most express the maturation markers 2A1, CD86 and CD40[59]. However, a small subset of Langerin-positive cells were found to be 2A1$^-$, and thus seemed to be immature[59]. As discussed below, these immature cells may be important in presenting pMHC class II complexes derived from soluble antigens acquired in the lymph nodes.

The sites at which the DCs in the aforementioned studies acquired antigen are not always apparent. It has been assumed that most of the DCs that present pMHC class II complexes to CD4 T cells acquired their antigens at the point of entry in the body before migrating to the lymph node, as studies of lymph node architecture indicate that lymph-borne soluble antigens do not have free access to the T cell areas[71].

Antigens injected subcutaneously are transported from the tissue into the subcapsular sinus of a connected lymph node through the afferent lymphatic vessels[72]. Narrow conduits connect the subcapsular sinus to perivenular spaces that surround the HEVs that pass through the lymph node[71]. These conduits are composed of type I and type III collagen fibers wrapped continuously on the outside with a coating of 90% reticular fibroblasts and 10% other cells, including DCs[71,73]. The lumen of each conduit is not completely filled with the fibers, as small molecules pass from the subcapsular sinus into the perivenular space through the conduits. This network of conduits allows soluble proteins generated in the tissue, such as cytokines and chemokines, to end up on the lumenal side of the HEV.

Although the conduits are abundant in the T cell areas in which naive T cells reside, large amounts of soluble molecules do not pass from the subcapsular sinus or conduits into this area[74]. However, two studies have indicated that leakage of soluble antigens out of the conduits probably occurs in some conditions[14,43]. Both studies showed that pMHC class II complexes were made by lymph node DCs by 3 hours after a subcutaneous injection of antigen. As tissue-derived DCs require about 12 hours to migrate to the lymph node[49], the DCs that produced these early pMHC class II complexes must have already been in the lymph node when they acquired the antigen.

The observation that pMHC class II complexes are generated at very early times indicates that at least some soluble antigen can leak out of the conduits. This idea is supported by a study of lymph node structure that showed collagen sheaths surrounding HEVs are not solid structures, but contain small pores[75]. If soluble antigens leak out of the conduits, possibly through these pores, then it is reasonable to assume that the cells nearest the conduits would have the best access to this source of antigen (**Supplementary Video** online, 00:00:24–00:00:31). Indeed, it was found that although most subcutaneously injected soluble antigen was present in the conduits at 30 minutes (**Fig. 2a**,**b**), some of the antigen was in CD40$^+$ cells near the conduits (**Fig. 2c**,**d**). These cells were CD11c$^+$CD11b$^+$ Langerhans cell immigrants and dermal DC immigrants[14]. The fact that these cells produced pMHC class II complexes from antigen that leaked from the conduits indicates that tissue-derived DCs are still capable of antigen uptake and processing even after migrating. This is unexpected, as post-migration DCs are thought to have undergone the maturation process, and thus should be inefficient at antigen uptake. However, one study indicated that this is not true for *in vivo*–matured DCs[76]. Langerhans cells that were induced to mature and migrate from the skin through the use of a skin irritant and lipopolysaccharide, and were then isolated from the lymph node, retained their capacity to endocytose proteins, compared with *in vitro*–matured Langerhans cells that had lost that ability.

The acquisition of free antigen from the conduits does not reduce the importance of migrating DCs. Even in the study in which evidence of presentation of free antigen was found[43], it is likely that tissue-resident DCs that acquired the antigen at the injection site migrated to the lymph node and presented pMHC class II complexes. Indeed, the CD11c$^+$CD8α^- cells identified as APCs in this study could have been such interstitial DC immigrants. Similarly, the interstitial DCs shown to migrate from the vaginal epithelium were probably the main presenters of herpes simplex virus 2 pMHC class II complexes[68]. Finally, it was shown that dermal DCs, which pick up subcutaneously injected antigen in the tissue and migrate to the lymph node at 24 hours, sustain the activation of CD4 T cells that were initially activated by pMHC class II complexes on resident lymph node DCs[14] (**Supplementary Video** online, 00:00:32–00:00:38).

Functional effects of DC antigen presentation

Although both resident and migrating DCs have the capacity to acquire antigens and generate pMHC class II complexes, differences between the two populations could have functional consequences Most evidence in the literature indicates that pMHC class II presentation by DCs that recently migrated from nonlymphoid tissue stimulates T cell immunity. This property may be explained by the fact that the migration of DCs from tissues is coupled with stabilization of pMHC complexes and increased expression of costimulatory molecules such as CD80 and CD86[44,69,77], which are required for optimal activation of naive T cells. Because the rate of DC migration into lymphoid organs from nonlymphoid tissues is low in the absence of inflammation[50,59], pMHC complex presentation by highly stimulatory immigrants should only predominate in conditions of tissue damage or infection, in which productive T cell immunity is desirable.

Although recent immigrants may be inherently stimulatory, other DCs in the secondary lymphoid organs can be tolerogenic, at least in certain circumstances. Mice injected with a DC-specific rat IgG2b monoclonal antibody developed rat IgG2b–specific T and B cell tolerance, whereas mice injected with this antibody plus an inflammatory adjuvant (IL-1) generated anti-rat IgG2b–specific immunity[78]. This type of antigen-targeting experiment was repeated[2], but with the important modification that antigen-specific CD4 T cells were monitored directly. CD205$^+$ DCs (blood-derived CD8α^+ DCs, Langerhans cells and interstitial DCs) were targeted *in vivo* for antigen uptake by injection of antibody to CD205 fused to a peptide from hen egg lysozyme (αDEC-HEL). HEL-specific naive CD4 T cells in these mice proliferated and then disappeared in mice injected with αDEC-HEL alone. In contrast, when an agonistic antibody to CD40 was injected with the αDEC-HEL, DCs expressed more CD40 and CD86, and the antigen-specific T cells proliferated and survived as memory cells. Therefore, resident lymph node DCs are capable of inducing T cell tolerance or immunity, depending on the inflammatory context in which they present pMHC complexes.

The finding that antigen targeting to DCs in the absence of inflammation is tolerogenic for T cells raises the possibility that DCs are involved in the induction of peripheral tolerance to self antigens. DCs may be normally kept in a tolerogenic state by a process involving engulfment of apoptotic cells. Mice with genetic defects in the disposal of apoptotic cells have constitutively activated DCs[79] and develop autoimmunity[79,80]. Disposal of apoptotic cells and maintenance of T cell tolerance may be linked by the fact that phagocytes that take up apoptotic cells produce anti-inflammatory mediators such as IL-10[81,82,83], which inhibit the expression of costimulatory molecules[84]. Resident DCs are well positioned for this purpose, as they constitutively express self pMHC complexes[85] and are located near naive T cells, a fraction of which must be apoptotic as a consequence of normal senescence. As migration of acutely activated DCs from nonlymphoid tissues is thought to require inflammation[51,57], these suppressed resident DCs would be the main cells capable of presenting pMHC complexes derived from self antigens or foreign antigens injected in the absence of inflammation.

Future directions

Most of our understanding of how DCs process and present antigens to T cells comes from *in vitro* studies. Although in some cases antigen is administered *in vivo*, *in vitro* T cell assays are still generally used to detect the presence of pMHC complexes. This process involves the removal of the DC and T cell from their natural environment, and thus results from such experiments must be viewed with caution. Other experiments attempting to identify antigen presentation events *in situ* have depended on differences in the frequency of interactions between T cells and DCs to determine when antigen presentation was occurring. Although this is a more direct approach, it still suffers from the limitation that the cells found interacting with antigen-specific T cells were not shown to display the relevant pMHC complexes. An ideal system would allow the real-time detection of pMHC complexes, using monoclonal antibodies[63,86–88]. Together with antibodies identifying antigen-specific T cells, it should be possible to directly observe antigen presentation *in vivo*.

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